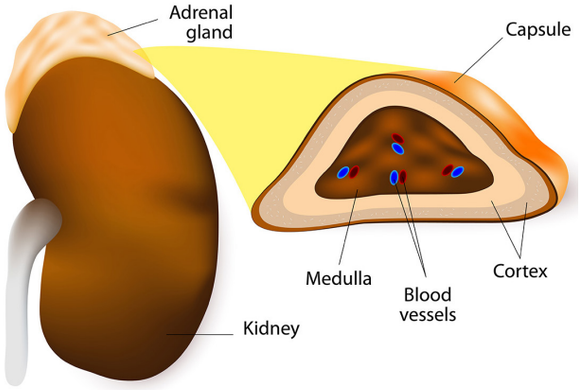



**Figure 16.11** Adrenal Gland

## Adrenal Cortex

The adrenal cortex consists of three layers of cells. They are **zona glomerulosa**, **zona fasciculata** and **zona reticularis**

### Hormones of Adrenal Cortex

The hormones secreted by the adrenal cortex are corticosteroids. They are classified into

a. Glucocorticoids
b. Mineralocorticoids

### Functions of adrenocortical hormones

#### Glucocorticoids

The glucocorticoids secreted by the zona fasciculata are **cortisol** and **corticosterone**

- They regulate cell metabolism.
- It stimulates the formation of glucose from glycogen in the liver.
- It is an anti-inflammatory and anti-allergic agent.

#### Mineralocorticoids

The mineralocorticoids secreted by zona glomerulosa is **aldosterone**

- It helps to reabsorb sodium ions from the renal tubules.
- It causes increased excretion of potassium ions.
- It regulates electrolyte balance, body fluid volume, osmotic pressure and blood pressure.

**More to Know**

The **cortisol** hormones of adrenal cortex serves to maintain the body in living condition and recover it from the severe effects of stress reactions. Thus an increased output of cortisol is "life saving" in "shock conditions". It is also known as life-saving hormone.

## Adrenal Medulla

The adrenal medulla is composed of **chromaffin cells**.They are richly supplied with sympathetic and parasympathetic nerves.

### Hormones of Adrenal Medulla

It secretes two hormones namely

a. Epinephrine (Adrenaline)
b. Norepinephrine (Noradrenaline)

They are together called as "**Emergency hormones**". It is produced during conditions of stress and emotion. Hence it is also referred as "flight, fright and fight hormone".

### Functions of adrenal medullary hormones

#### Epinephrine (Adrenaline)

- It promotes the conversion of glycogen to glucose in liver and muscles.
- It increases heart beat and blood pressure.
- It increases the rate of respiration by dilation of bronchi and trachea.
- It causes dilation of the pupil in eye.
- It decreases blood flow through the skin.

#### Norepinephrine (Noradrenalin)

Most of its actions are similar to those of epinephrine.

## 16.2.6 Reproductive Glands (Gonads)

The sex glands are of two types the **testes** and the **ovaries**. The testes are present in male, while the ovaries are present in female.



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